



PROJECT NAME: SECUREDRIIVE

VERSION: 1.1

DESCRIPTION

SecureDrive is an open-source endpoint security tool for users. It will scan the device for any virus or harmful files and provide real-time scan capabilities. Users will also have the functionality to check if their email has been compromised in a data breach and if their password has been exposed in a data breach.

PROJECT SCOPE

The scope of this project includes the development of an antivirus software tool that provides comprehensive security features for end users. The primary functions will involve scanning for viruses and harmful files, real-time scanning of new files, and checking for email and password breaches.

OBJECTIVES

- Develop a system scanning feature that scans the entire device for malicious files.
- Implement a file scanning feature that allows users to scan specific files for malware.
- Create a real-time scanning feature that monitors new downloads and alerts users if any malicious files are detected.
- Integrate an email breach checking functionality that verifies if the user's email has been part of any known data breaches.
- Integrate a password breach checking functionality that verifies if the user's password has been part of any known data breaches.

GOALS

- To provide a user-friendly, reliable, and efficient security tool.
- To maintain the highest standards of security and privacy for users' data.

- To foster a community-driven approach through open-source development and collaboration.

FUNCTIONAL REQUIREMENTS

● System Scan

- The software will scan the entire system for any malicious or harmful files.
- Users can initiate a full system scan manually.
- The program will scan the hash of the files in the system against our database of malicious hashes.
- The scan will provide a detailed report of findings and actions taken.
- If a malicious file is detected, the antivirus will quarantine it and will report it to the user.

● File Scan

- Users can select and scan specific files.
- The program will scan the hash of the file against our database of malicious hashes.
- The software will analyze the file and provide an immediate report on whether it is malicious.
- The system will scan the hash of the file against our database of malicious Hashes.

● Real-Time Scan

- The software will automatically scan files as they are downloaded or created on the system.
- The program will scan the hash of the new downloaded file against our database of malicious hashes.
- It will alert users immediately if any malicious file is detected.
- If a malicious file is detected, the antivirus will quarantine it and will report it to the user.

● Email Breach Checker:

- Users can enter their email addresses to check if they have been part of any known data breaches.
- Results will be displayed to the user with recommendations for action.



- **Password Breach Checker**

- Users can enter their passwords to check if they have been exposed in any known data breaches.
- Results will be displayed to the user with recommendations for action.

NON-FUNCTIONAL REQUIREMENTS

- **Performance**

- The software should perform scans efficiently without significantly impacting system performance.
- Real-time scanning should have minimal latency to ensure user activities are not disrupted.
- Scanning processes should be fast and quick with no false positives or false negatives.
- It should automatically fetch the third-party hash database, to update the new hash values that could be malicious, and update the local database accordingly..

- **Security**

- All user data, including emails and passwords, must be securely handled and encrypted.
- SecureDrive will not collect any personal data; all data used for verification will most likely be handled by VirusTotal.

- **Usability**

- The user interface should be intuitive and easy to navigate.
- Clear instructions and feedback should be provided for each feature.

- **Reliability**

- The software should be reliable and stable, with minimal crashes or bugs.
- It should handle errors gracefully and provide meaningful error messages to the user.
- It should automatically fetch the third-party hash database, to update the new hash values that could be malicious, and update the local database accordingly.



- **Maintainability**

- The codebase should be well-documented to facilitate maintenance and updates.
- Comments should be written to ensure the code is easier to understand and document.
- The software should be designed with modularity to allow for easy addition of new features and improvements in newer versions of SecureDrive.

TECHNOLOGIES

- Backend Language: To Be Determined (TBD)
- Frontend Language: To Be Determined (TBD)
- Code Deployment: GitHub
- Communication Tools: Slack, WhatsApp
- Collaboration Tools: Google Docs, Google Sheets, Google Drive

DEVELOPMENT AND DEPLOYMENT

- **Development**

The project will follow Agile methodology with iterative development cycles. The development process will include regular sprint planning, daily stand-ups, sprint reviews, and retrospectives.

- **Deployment**

- The code will be hosted on GitHub for version control and collaboration.
- Automated testing and continuous integration will be used to ensure code quality and stability.
- Regular updates will be deployed to keep the software current with the latest security definitions and features.